

Five–Pillar Fundamental Scoring Framework

Complete mathematical formulations, accounting data inputs, scoring bands, and financial rationale

Pocket Omaha: Five–Pillar Fundamental Scoring Framework

This document provides a comprehensive technical and financial specification for the **Five–Pillar Scoring Engine** in Pocket Omaha ([core/scoring.js](#)), which powers the **Pillar Comparison Radar Chart** in the peer analysis screen ([web/app.js](#)).

1. Core Architecture & Philosophy

The scoring engine is built on five core principles:

- 1. **No Artificial Fallbacks:** A metric that cannot be computed from filed numbers is reported as `unavailable` . It never defaults to an arbitrary constant (e.g., 0 or average), preventing false confidence.
- 2. **Dynamic Denominator Adjustment:** Inapplicable or unavailable sub-metrics are omitted from both the numerator (earned points) and denominator (possible points). A bank is not penalized for lacking a gross profit line or inventory.
- 3. **Pillar Normalization:** Every pillar is standardized to a 0 to 20 point scale:

$$\text{Pillar Score} = \left(\frac{\sum \text{Earned Points}}{\sum \text{Measurable Possible Points}} \right) \times 20$$

- 4. **Coverage Guardrails:** If fewer than 60% of total items can be measured ($\text{Coverage} < 0.60$), no overall composite health score is emitted.
- 5. **Radar Geometry:** For each axis $i \in \{0, 1, 2, 3, 4\}$, the radial distance from the origin on a unit scale $[0, 1]$ is:

$$\text{Fraction}_i = \frac{\text{Pillar Score}_i}{20}$$

If a pillar is unmeasured, its coordinate collapses to the center (0) rather than projecting an assumed midpoint.

2. Axis Breakdown: Formulas, Inputs & Financial Reasoning

1. Solvency 20 pts - Altman Z-Score (or Equity/Assets) - Net Debt / EBITDA - Interest Coverage Ratio - Current & Quick Ratios	2. Profitability 20 pts - Piotroski F-Score (9-pt) - ROIC (or ROE for Financials) - Operating Margin Trend (bps) - Free Cash Flow Conversion	3. Valuation 20 pts - Forward P/E vs 5Y Median - PEG Ratio (growth-adjusted) - EV / Free Cash Flow Yield - Discount to 2-Stage DCF	4. Growth 20 pts - Multi-Year Revenue CAGR - Diluted EPS CAGR - FCF per Share CAGR - Gross Margin Trajectory	5. Capital 20 pts - Buybacks vs Dilution (%/yr) - Dividend Safety / Reinvestment - Asset Turnover Efficiency
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Pillar 1: Solvency (*Financial Health & Solvency*) — Top Spoke

Objective: Gauge bankruptcy vulnerability, structural leverage, interest service capability, and short-term liquidity.

1.1 Altman Z-Score (Non-Financials)

- **Mathematical Formula:**

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

Where:

- $X_1 = \frac{\text{Working Capital}}{\text{Total Assets}} = \frac{\text{Current Assets} - \text{Current Liabilities}}{\text{Total Assets}}$
- $X_2 = \frac{\text{Retained Earnings}}{\text{Total Assets}}$
- $X_3 = \frac{\text{EBIT}}{\text{Total Assets}}$
- $X_4 = \frac{\text{Market Capitalization}}{\text{Total Liabilities}}$
- $X_5 = \frac{\text{Total Revenue}}{\text{Total Assets}}$
- **Data Inputs:** Balance sheet (Assets, Liabilities, Current Assets/Liabilities, Retained Earnings), Income Statement (EBIT, Revenue), Market Cap.
- **Scoring Bands (Max 5 pts):**
 - $Z \geq 3.0$ (Safe Zone): **5.0 pts**
 - $1.8 \leq Z < 3.0$ (Grey Zone): **3.0 pts**
 - $Z < 1.8$ (Distress Zone): **0.0 pts**
- **Financial Reasoning:** Edward Altman's 1968 model predicts insolvency probability within 2 years. Weightings penalize high liabilities relative to market equity and low operating earnings on assets.
- **Sector Adaptation:** Inapplicable to Financials/Banks (where balance sheets have no working-capital cycle).

1.2 Net Debt / EBITDA (Non-Financials)

- **Mathematical Formula:**

$$\text{Net Cash} = (\text{Cash} + \text{Short-Term Investments}) - \text{Total Debt}$$

$$\text{Net Debt} / \text{EBITDA} = \frac{-\text{Net Cash}}{\text{EBITDA}} = \frac{\text{Total Debt} - \text{Cash \& Equiv}}{\text{EBITDA}}$$

- **Scoring Bands (Max 5 pts):**
 - Net Cash > 0 (Net cash balance / negative net debt): **5.0 pts**
 - Net Debt / EBITDA $\leq 1.5\times$: **4.0 pts**
 - $1.5\times < \text{Net Debt} / \text{EBITDA} \leq 3.0\times$: **2.0 pts**
 - Net Debt / EBITDA $> 3.0\times$: **0.0 pts**
- **Financial Reasoning:** Measures how many years of current operating cash earnings would be required to pay off all debt net of available cash. A net cash position provides an unassailable balance sheet cushion.

1.3 Interest Coverage (Non-Financials)

- **Mathematical Formula:**

$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

- **Scoring Bands (Max 5 pts):**
 - Unburdened (Total Debt = 0 or Interest Expense = 0): **5.0 pts**
 - $\geq 8.0\times$: **5.0 pts**
 - $\geq 4.0\times$: **3.0 pts**
 - $\geq 1.5\times$: **1.0 pt**
 - $< 1.5\times$: **0.0 pts**
- **Financial Reasoning:** High coverage guarantees that an operational downturn will not trigger technical default on debt covenants.

1.4 Current & Quick Ratios (Non-Financials)

- **Mathematical Formula:**

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}, \quad \text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}}$$

- **Scoring Bands (Max 5 pts):**
 - Current Ratio ≥ 1.5 AND Quick Ratio ≥ 1.0 : **5.0 pts**
 - Current Ratio ≥ 1.5 : **4.0 pts**
 - Current Ratio ≥ 1.0 : **3.0 pts**
 - Current Ratio < 1.0 : **0.0 pts**

- **Financial Reasoning:** Assesses immediate 12-month liquidity. Subtracting inventory prevents distorted liquidity figures from slow-moving stock.

1.5 Equity to Assets (Financials Only)

- **Mathematical Formula:**

$$\text{Equity to Assets \%} = \frac{\text{Total Equity}}{\text{Total Assets}} \times 100$$

- **Scoring Bands (Max 5 pts, rescaled to 20):**
 - $\geq 10.0\%$: **5.0 pts** | $\geq 8.0\%$: **4.0 pts** | $\geq 6.0\%$: **2.5 pts** | $< 6.0\%$: **0.0 pts**
- **Financial Reasoning:** The standard regulatory leverage indicator reflecting bank capital adequacy and loss-absorption capacity.

Pillar 2: Profitability (*Profitability & Moat Quality*) — Top-Right Spoke

Objective: Measure operational excellence, economic moat durability, accrual quality, and cash generation capability.

2.1 Piotroski F-Score

- **Mathematical Formula:** Sum of 9 binary tests ($F \in [0, 9]$):
 1. Net Income > 0
 2. Operating Cash Flow (OCF) > 0
 3. $\Delta\text{ROA} > 0 \implies \frac{\text{Net Income}_t}{\text{Assets}_t} > \frac{\text{Net Income}_{t-1}}{\text{Assets}_{t-1}}$
 4. Accrual Quality $\implies \text{OCF}_t > \text{Net Income}_t$
 5. $\Delta\text{Leverage} \leq 0 \implies \frac{\text{LTD}_t}{\text{Assets}_t} \leq \frac{\text{LTD}_{t-1}}{\text{Assets}_{t-1}}$
 6. $\Delta\text{Current Ratio} > 0 \implies \text{CR}_t > \text{CR}_{t-1}$
 7. No Dilution $\implies \text{Diluted Shares}_t \leq \text{Diluted Shares}_{t-1}$
 8. $\Delta\text{Gross Margin} > 0 \implies \frac{\text{Gross Profit}_t}{\text{Revenue}_t} > \frac{\text{Gross Profit}_{t-1}}{\text{Revenue}_{t-1}}$
 9. $\Delta\text{Asset Turnover} > 0 \implies \frac{\text{Revenue}_t}{\text{Assets}_t} > \frac{\text{Revenue}_{t-1}}{\text{Assets}_{t-1}}$
- **Normalization:** If fewer than 9 tests are testable (min 6), normalized as $\text{round}\left(\frac{\text{Score}}{\text{Testable}} \times 9\right)$.
- **Scoring Bands (Max 5 pts):**
 - $F \geq 8$: **5.0 pts** | $F \geq 6$: **3.5 pts** | $F \geq 4$: **2.0 pts** | $F < 4$: **0.0 pts**
- **Financial Reasoning:** Joseph Piotroski's accounting screen separates improving fundamental momentum from accounting deterioration.

2.2 ROIC (Non-Financials) & ROE (Financials)

- **Mathematical Formula:**

$$\text{ROIC \%} = \frac{\text{EBIT} \times (1 - t)}{\text{Total Debt} + \text{Total Equity} - \text{Cash}} \times 100$$

(where t = Effective Tax Rate, clamped to statutory rate if loss-making)

$$\text{ROE \% (Financials)} = \frac{\text{Net Income}}{\text{Total Equity}} \times 100$$

- **Special Handling:** If Invested Capital ≤ 0 (e.g. negative book equity due to extensive buybacks), ROIC is undefined and reported as `null` rather than a falsely astronomical figure.
- **Scoring Bands (Max 5 pts):**
 - $\geq 15.0\%$: **5.0 pts** | $\geq 10.0\%$: **3.5 pts** | $\geq 5.0\%$ (or $\geq 6.0\%$ for ROE): **2.0 pts** | Below: **0.0 pts**
- **Financial Reasoning:** True proxy of economic moat. A company earning above 15% ROIC generates substantial economic value over its weighted cost of capital (WACC).

2.3 Operating Margin Trend (Non-Financials)

- **Mathematical Formula:**

$$\Delta \text{OpMargin (bps)} = \left(\frac{\text{Operating Income}_t}{\text{Revenue}_t} - \frac{\text{Operating Income}_{t-1}}{\text{Revenue}_{t-1}} \right) \times 10,000$$

- **Scoring Bands (Max 5 pts):**
 - $\geq +100$ bps (+1.0% expansion): **5.0 pts**
 - ≥ -50 bps (stable): **3.5 pts**
 - ≥ -200 bps: **1.5 pts**
 - < -200 bps: **0.0 pts**
- **Financial Reasoning:** Focuses on margin direction rather than static level; expanding margins demonstrate pricing power and operating leverage.

2.4 Free Cash Flow Conversion

- **Mathematical Formula:**

$$\text{FCF Conversion \%} = \frac{\text{Free Cash Flow}}{\text{Net Income}} \times 100 = \frac{\text{Operating Cash Flow} - \text{CapEx}}{\text{Net Income}} \times 100$$

- **Scoring Bands (Max 5 pts):**
 - $\geq 100\%$ (Owner earnings exceed GAAP profit): **5.0 pts**
 - $\geq 80\%$: **3.5 pts**
 - $\geq 50\%$: **2.0 pts**
 - $< 50\%$: **0.0 pts**
- **Financial Reasoning:** High conversion confirms accounting earnings are backed by real liquid cash flow rather than aggressive accruals.

Pillar 3: Valuation (*Valuation & Margin of Safety*) — Bottom-Right Spoke

Objective: Determine whether market price provides an adequate margin of safety relative to intrinsic cash flow yield and historical multiples.

3.1 Forward P/E vs History / Absolute Multiple

- **Mathematical Formula:**

$$\% \text{ vs 5Y Median} = \frac{\text{Forward P/E} - \text{5Y Median P/E}}{\text{5Y Median P/E}} \times 100$$

- **Scoring Bands (Max 5 pts):**
 - *When 5Y history is present:*
 - $\geq 15\%$ discount: **5.0 pts**
 - $\leq 10\%$ premium: **3.5 pts**
 - $\leq 25\%$ premium: **2.0 pts**
 - $> 25\%$ premium: **0.0 pts**
 - *Fallback to absolute Forward P/E:*
 - $\leq 18.0\times$: **5.0 pts** | $\leq 26.0\times$: **3.5 pts** | $\leq 35.0\times$: **2.0 pts** | $> 35.0\times$: **0.0 pts**
- **Financial Reasoning:** Prevents punishing structural compounders for trading at higher absolute multiples if they trade at a discount to their historical valuation band.

3.2 PEG Ratio (Price / Earnings to Growth)

- **Mathematical Formula:**

$$\text{PEG} = \frac{\text{Forward P/E}}{\text{Expected Annual EPS Growth Rate \%}}$$

- **Special Rule:** If $\text{PEG} \leq 0$ (negative growth), it is marked **unavailable** (0 denominator weight) because shrinking earnings reflect fundamental risk, not value.
- **Scoring Bands (Max 5 pts):**
 - $\text{PEG} \leq 1.0\times$: **5.0 pts**

- $PEG \leq 1.8\times$: **3.5 pts**
- $PEG \leq 2.5\times$: **2.0 pts**
- $PEG > 2.5\times$: **0.0 pts**
- **Financial Reasoning:** Peter Lynch's valuation metric adjusting P/E for earnings compounding pace.

3.3 EV / Free Cash Flow Yield

- **Mathematical Formula:**

Enterprise Value (EV) = Market Cap + Total Debt – Cash & Short-term Investments

$$EV / FCF \text{ Yield \%} = \frac{\text{Free Cash Flow}}{\text{Enterprise Value}} \times 100$$

- **Scoring Bands (Max 5 pts):**
 - $\geq 6.0\%$: **5.0 pts** | $\geq 4.0\%$: **3.5 pts** | $\geq 2.0\%$: **2.0 pts** | $< 2.0\%$: **0.0 pts**
- **Financial Reasoning:** Capital-structure-neutral cash flow yield. Accounts for both equity valuation and net debt burden.

3.4 Discount to Two-Stage DCF Fair Value

- **Mathematical Formula:**

$$\text{Discount to Fair Value \%} = \frac{\text{DCF Fair Value per Share} - \text{Current Price}}{\text{DCF Fair Value per Share}} \times 100$$

- **DCF Engine Highlights:**
 - **Cash Flow Base:** Uses 3-year median if latest FCF is an outlier ($> 1.35\times$ or $< 0.65\times$ median).
 - **Growth Rate (g):** Median of filed CAGRs, capped at Revenue CAGR + 5% (max 20%). Negative growth is projected downwards if latest revenue shrank.
 - **Terminal Multiple:** Dynamically derived from business quality (ROIC, Gross Margin, Net Cash) between $8\times$ and $26\times$.
 - **Discount Rate:** 9.5%.
- **Scoring Bands (Max 5 pts):**
 - $\geq +20\%$ Discount (Margin of safety): **5.0 pts**
 - $\geq -10\%$ (Within 10% premium): **3.0 pts**
 - $\geq -20\%$ (Within 20% premium): **1.0 pt**
 - Over 20% premium: **0.0 pts**

Pillar 4: Growth (*Growth & Operating Leverage*) — Bottom-Left Spoke

Objective: Assess top-line expansion, bottom-line compounding, per-share value creation, and pricing trajectory.

4.1 Revenue CAGR

- **Mathematical Formula:**

$$\text{Revenue CAGR} = \left(\frac{\text{Revenue}_t}{\text{Revenue}_{t-n}} \right)^{\frac{1}{n}} - 1$$

- **Scoring Bands (Max 5 pts):**
 - $\geq 15.0\%$: **5.0 pts** | $\geq 8.0\%$: **3.5 pts** | $\geq 3.0\%$: **2.0 pts** | $< 3.0\%$: **0.0 pts**
- **Financial Reasoning:** Top-line revenue expansion proves product-market fit and total addressable market demand.

4.2 Diluted EPS CAGR

- **Mathematical Formula:**

$$\text{EPS CAGR} = \left(\frac{\text{Diluted EPS}_t}{\text{Diluted EPS}_{t-n}} \right)^{\frac{1}{n}} - 1$$

- **Scoring Bands (Max 5 pts):**
 - $\geq 20.0\%$: **5.0 pts** | $\geq 10.0\%$: **3.5 pts** | $\geq 0.0\%$: **1.5 pts** | $< 0.0\%$: **0.0 pts**

- **Financial Reasoning:** Compounding earnings per share reflects true per-share shareholder accretion.

4.3 FCF per Share CAGR (Non-Financials)

- **Mathematical Formula:**

$$\text{FCF per Share CAGR} = \left(\frac{\text{FCF} / \text{Share}_t}{\text{FCF} / \text{Share}_{t-n}} \right)^{\frac{1}{n}} - 1$$

- **Scoring Bands (Max 5 pts):**
 - $\geq 15.0\%$: **5.0 pts** | $\geq 5.0\%$: **3.5 pts** | $\geq 0.0\%$: **1.5 pts** | $< 0.0\%$: **0.0 pts**
- **Financial Reasoning:** Ensures bottom-line growth is translating into cash available for reinvestment or capital return.

4.4 Gross Margin Trajectory (Non-Financials)

- **Mathematical Formula:** Quarterly delta over filed quarters (or YoY basis points):

$$\Delta \text{Gross Margin (bps)} = (\text{Gross Margin}_{\text{latest}} - \text{Gross Margin}_{\text{base}}) \times 10,000$$

- **Scoring Bands (Max 5 pts):**
 - ≥ 0 bps (Steady or expanding): **5.0 pts**
 - ≥ -100 bps: **3.5 pts**
 - ≥ -300 bps: **1.5 pts**
 - < -300 bps: **0.0 pts**
- **Financial Reasoning:** Gross margin is where pricing power and inflation resistance manifest first.

Pillar 5: Capital (*Capital Allocation & Returns*) — Top-Left Spoke

Objective: Evaluate management stewardship in share buybacks, dividend sustainability, and balance sheet asset productivity.

5.1 Buybacks vs. Dilution

- **Mathematical Formula:**

$$\Delta \text{Shares}_{\text{annualized}} = \left(\left(1 + \frac{\text{Shares}_t - \text{Shares}_{t-n}}{\text{Shares}_{t-n}} \right)^{\frac{1}{n}} - 1 \right) \times 100$$

- **Scoring Bands (Max 7 pts):**
 - $\leq -2.0\%/yr$ (Active accretive buybacks): **7.0 pts**
 - $\leq +0.5\%/yr$ (Negligible dilution / organic creep): **5.0 pts**
 - $\leq +2.0\%/yr$ (Moderate stock-based comp dilution): **2.0 pts**
 - $> +2.0\%/yr$ (Heavy dilution): **0.0 pts**
- **Financial Reasoning:** Accretive buybacks permanently enhance per-share value; aggressive stock-based compensation quietly transfers enterprise value away from public shareholders.

5.2 Dividend Safety OR Reinvestment Quality

- **Mathematical Logic:**
 - **Case A: Dividend Payer** (Dividend Yield > 0):

$$\text{FCF Dividend Payout Ratio} = \frac{\text{Dividends Paid}}{\text{Free Cash Flow}}$$

- Payout $< 60\%$ AND ≥ 5 -year streak: **7.0 pts**
- Payout $< 60\%$: **5.5 pts**
- Payout $< 90\%$: **3.0 pts**
- Payout $\geq 90\%$: **1.0 pt**
- **Case B: Non-Dividend Payer** (Scored on Reinvestment Efficiency via ROIC):
 - ROIC $\geq 15.0\%$: **7.0 pts**
 - ROIC $\geq 10.0\%$: **5.0 pts**

- ROIC $\geq$ 5.0%: **3.0 pts**
- ROIC $<$ 5.0%: **0.0 pts**
- **Financial Reasoning:** Retaining 100% of earnings is only rational if management reinvests capital at high incremental returns (ROIC $\geq$ 15%). If dividends are paid, they must be safely covered by free cash flow.

5.3 Asset Turnover Efficiency

- **Mathematical Formula:**

$$\text{Asset Turnover} = \frac{\text{Total Revenue}}{\text{Total Assets}}$$

- **Scoring Bands (Max 6 pts):**
 - *When Sector Peer Median is available:*
 - $\geq 1.25 \times$ Sector Median: **6.0 pts**
 - $\geq 0.90 \times$ Sector Median: **4.5 pts**
 - $\geq 0.60 \times$ Sector Median: **3.0 pts**
 - $< 0.60 \times$ Sector Median: **0.0 pts**
 - *Absolute Fallback:*
 - $\geq 0.80 \times$: **6.0 pts** | $\geq 0.40 \times$: **4.0 pts** | $< 0.40 \times$: **0.0 pts**
- **Financial Reasoning:** Compares revenue generated per dollar of asset base against industry peers to distinguish asset-light, highly efficient business models.

3. Pillar Radar Chart Geometry & Rendering

In [web/app.js](#), the SVG radar polygon is generated across the 5 spokes:

```
const AXES = ['Solvency', 'Profitability', 'Valuation', 'Growth', 'Capital'];
const size = 260;
const c = size / 2; // Center (130, 130)
const rMax = c - 34; // Max radius = 96px

const point = (axisIndex, fraction) => {
  // Rotate by -90 deg so axis 0 starts at 12 o'clock
  const angle = (Math.PI * 2 * axisIndex) / AXES.length - Math.PI / 2;
  const r = rMax * Math.max(0, Math.min(1, fraction));
  return [c + r * Math.cos(angle), c + r * Math.sin(angle)];
};
```

Axis Index	Spoke Name	Angle	Polar Position on Radar
0	Solvency	-90°	Top (12 o'clock)
1	Profitability	-18°	Top-Right
2	Valuation	+54°	Bottom-Right
3	Growth	+126°	Bottom-Left
4	Capital	+198°	Top-Left

Concentric web polygon rings are drawn at **25%** (5/20), **50%** (10/20), **75%** (15/20), and **100%** (20/20).

4. Glossary of Financial & Technical Terms

Term	Definition & Application in Pocket Omaha
Altman Z-Score	A credit-strength formula published by Edward Altman (1968) that combines five financial ratios to predict the probability of bankruptcy within two years. A score ≥ 3.0 indicates low distress risk.
Asset Turnover	$\frac{\text{Revenue}}{\text{Total Assets}}$. Measures the efficiency with which a company uses its assets to generate top-line sales.
Basis Points (bps)	A unit of measure equal to one-hundredth of a percentage point (1 bps = 0.01% = 0.0001). Used to express changes in margins (e.g., +100 bps = +1.0%).
Beta (β)	A measure of stock price volatility and systematic risk relative to the broader market. Used in CAPM to estimate the cost of equity. Clamped in Pocket Omaha to [0.6, 2.5] to prevent distorted cost-of-capital estimates.
Book Equity (Book Value)	Total Assets – Total Liabilities. The net accounting net worth of the company. Can become negative when cumulative share repurchases exceed retained earnings.
CAGR (Compound Annual Growth Rate)	The annualized geometric mean growth rate of a financial metric over a specified multi-year period n : $\left(\frac{\text{End}}{\text{Start}}\right)^{1/n} - 1$.
CapEx (Capital Expenditures)	Funds used by a company to acquire, upgrade, and maintain physical assets such as property, plant, buildings, technology, or equipment.
CAPM (Capital Asset Pricing Model)	An equilibrium asset-pricing framework estimating the required return on equity: $\text{Cost of Equity} = R_f + \beta \times \text{ERP}$.
Coverage Threshold	Minimum fraction of measurable data items required (60%) for Pocket Omaha to emit an overall Health Score; prevents issuing misleading composite ratings for sparse data profiles.
Current Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$. A baseline measure of short-term liquidity indicating whether short-term obligations can be covered by short-term assets.
DCF (Discounted Cash Flow)	A valuation methodology that discounts projected future free cash flows to the present using a discount rate (9.5%), calculating intrinsic fair value per share.
Diluted Shares	The total number of shares that would be outstanding if all convertible securities (options, RSUs, warrants, convertible bonds) were exercised.
EBIT	<i>Earnings Before Interest and Taxes</i> . A company's operating earnings before financing costs and taxes.
EBITDA	<i>Earnings Before Interest, Taxes, Depreciation, and Amortization</i> . An approximate indicator of recurring pre-tax operational cash generation before non-cash capital depreciation charges.
Effective Tax Rate	$\frac{\text{Tax Provision}}{\text{Pretax Income}}$. The actual rate of tax paid on earnings. Substituted with statutory 21% when GAAP rate is negative or distorted by one-off tax credits.
Enterprise Value (EV)	Market Cap + Total Debt – Cash & Short-Term Investments. The theoretical total takeover cost of the firm, neutral to capital structure.
Equity Risk Premium (ERP)	The excess return that investing in the stock market provides over the risk-free rate (5.0% baseline in the WACC model).
Equity-to-Assets	$\frac{\text{Total Equity}}{\text{Total Assets}} \times 100$. Primary solvency and leverage metric used for banks and financial institutions in lieu of working capital / debt ratios.
FCF (Free Cash Flow)	Operating Cash Flow – CapEx. The surplus discretionary cash generated by the business that can be returned to shareholders (dividends, buybacks) or used to pay down debt without harming core operations.
FCF Conversion	$\frac{\text{Free Cash Flow}}{\text{Net Income}} \times 100$. Gauges earnings quality by measuring what percentage of accounting net income translates into actual liquid cash.

Term	Definition & Application in Pocket Omaha
FCF Yield	$\frac{\text{Free Cash Flow}}{\text{Market Cap}}$ or $\frac{\text{Free Cash Flow}}{\text{Enterprise Value}}$. The unlevered or levered cash return generated per dollar of company valuation.
Forward P/E	$\frac{\text{Current Price}}{\text{Consensus Next 12 Months EPS}}$. Valuation multiple based on forward expected earnings.
Gross Margin	$\frac{\text{Gross Profit}}{\text{Revenue}} \times 100$. Measures the direct profit generated after deducting cost of goods sold (COGS), serving as a prime indicator of pricing power and product differentiation.
Interest Coverage Ratio	$\frac{\text{EBIT}}{\text{Interest Expense}}$. Number of times operating profit covers contractual debt interest obligations.
Invested Capital	(Total Debt + Total Equity) – Cash & Cash Equivalents. The total net capital actively committed to operating assets.
Margin of Safety	The percentage discount between the intrinsic DCF fair value and the market share price, providing downside protection against projection errors.
Net Cash / Net Debt	(Cash + Short-Term Investments) – Total Debt. A positive figure indicates net cash surplus; a negative figure indicates net debt liability.
Net Debt / EBITDA	$\frac{\text{Total Debt} - \text{Cash}}{\text{EBITDA}}$. A leverage multiple representing the years of operating earnings needed to extinguish net debt.
NOPAT	<i>Net Operating Profit After Tax</i> (EBIT $\times$ (1 – Tax Rate)). Operating profit attributable to all capital providers after accounting for taxes.
Operating Cash Flow (OCF)	Cash generated from core day-to-day revenue-producing business activities, before capital expenditures.
Operating Leverage	The degree to which revenue growth translates into faster operating profit expansion due to fixed cost structures.
Operating Margin	$\frac{\text{Operating Income}}{\text{Revenue}} \times 100$. Percentage of revenue remaining after paying for variable and fixed operating expenses.
PEG Ratio	$\frac{\text{Forward P/E}}{\text{Annual Earnings Growth Rate \%}}$. Relates the price-to-earnings multiple directly to the underlying growth pace of the enterprise.
Piotroski F-Score	A discrete 9-point fundamental accounting score created by Joseph Piotroski evaluating profitability, leverage/liquidity, and operating efficiency.
Quick Ratio (Acid-Test)	$\frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}}$. A stringent liquidity test that excludes inventory from short-term assets.
Risk-Free Rate (R_f)	The baseline theoretical return of an investment with zero default risk (4.2% baseline representing 10-year US Treasuries).
ROA (Return on Assets)	$\frac{\text{Net Income}}{\text{Total Assets}} \times 100$. Efficiency of the overall balance sheet in generating net profit.
ROE (Return on Equity)	$\frac{\text{Net Income}}{\text{Shareholders' Equity}} \times 100$. Net profitability generated per dollar of accounting book equity.
ROIC (Return on Invested Capital)	$\frac{\text{NOPAT}}{\text{Invested Capital}} \times 100$. The gold standard for measuring economic moat strength and capital efficiency.
ROIC Spread	ROIC – WACC. The excess return earned over the firm's cost of capital. A positive spread proves value creation; a negative spread indicates capital destruction.
Terminal Multiple	An exit multiple applied to year-5 cash flows in the DCF model, dynamically adjusted ($8 \times$ to $26 \times$) based on the company's ROIC, margins, and balance sheet strength.
Terminal Value	The estimated present value of all cash flows beyond the explicit 5-year DCF projection period.
Trailing P/E	$\frac{\text{Current Price}}{\text{Trailing 12-Month Diluted EPS}}$. Valuation multiple based on actual historical filed earnings.
WACC (Weighted Average Cost of Capital)	The blended hurdle rate combining the cost of equity (via CAPM) and the after-tax cost of debt, weighted by their market values.

Term	Definition & Application in Pocket Omaha
Working Capital	Current Assets – Current Liabilities. Capital available for conducting daily business operations and meeting short-term obligations.